WORKSHEET

Chapter 10: Sample proportion confidence intervals

1 CAS Out of 56 people at a shopping centre in Melbourne, 38 had had a flu injection this flu season.

a Estimate the probability of someone in Melbourne having had a flu injection this flu season.

b Estimate the standard deviation of the sampling distribution.

c Estimate the 90% confidence interval for the probability of someone in Melbourne having had a flu injection this flu season.

d What does the confidence interval mean in this case?

2 CAS Out of 32 in the city mall in Brisbane, 14 were regular users of Twitter.

a Estimate the probability of someone in Brisbane being a regular Twitter user.

b Estimate the standard deviation of the sampling distribution.

c Estimate the 99% confidence interval for the probability of someone in Brisbane being a regular Twitter user.

d What does the confidence interval mean in this case?

3 CAS Find the approximate margin of error for each of the following confidence levels for the stated values of $\hat{p}$, n and a.

a $\hat{p} = 0.9, n = 50, a = 95$

b $\hat{p} = 0.6, n = 80, a = 90$

© Cengage Learning Australia Pty Ltd 2014

c $\hat{p} = 0.35, n = 120, a = 85$

d $\hat{p} = 0.75, n = 150, a = 99$

4 CAS Find the confidence intervals for each of the stated values of $\hat{p}$, n and a.

a $\hat{p} = 0.3, n = 50, a = 95$

b $\hat{p} = 0.85, n = 70, a = 85$

c $\hat{p} = 0.44, n = 35, a = 90$

d $\hat{p} = 0.72, n = 65, a = 98$

5 A sample of 50 has a sample proportion of 0.7. Find each of the following.

a The margin of error for a confidence level of 90% for the corresponding probability.

b The 90% confidence interval for the probability.

© Cengage Learning Australia Pty Ltd 2014

Answers

1 **a** $\frac{19}{28} \approx 0.68$

b About 0.062

c (0.58, 0.78)

d The probability of someone in Melbourne having had a flu injection this flu season has a 90% chance of lying in the interval from 0.58 and 0.78 (between 58% and 78%).

2 **a** $\frac{7}{16} = 0.4375$

b About 0.088

c (0.21, 0.66)

d The probability of someone in Brisbane being a regular Twitter user has a 99% chance of lying in the interval from 0.21 and 0.66 (between 21% and 66%).

3 **a** About 0.083

b About 0.090

c About 0.063

d About 0.091

4 **a** (0.17, 0.43)

b (0.79, 0.91)

c (0.30, 0.58)

d (0.59, 0.85)

5 **a** The margin of error is about 0.11.

b The 90% confidence interval is (0.59, 0.81).

© Cengage Learning Australia Pty Ltd 2014